PARTH PAREKH

linkedin.com/in/parekh422 | (973) 932-6070 | pvparekh14@gmail.com | github.com/pvparekh | parthparekh.dev

EDUCATION

Rutgers University – New Brunswick, NJ
B.S. in Computer Science | Minor in Data Science | May 2026
Relevant Coursework: Intro to Artificial Intelligence, Design and Analysis of Computer Algorithms, Data Structures, Software Methodology, Systems Programming, Computer Architecture, Data Management, Machine Learning Principles

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, R, HTML/CSS
Frontend: React, Next.js 15, Tailwind CSS, Framer Motion, Recharts
Backend: FastAPI, Node.js, REST APIs, WebSockets, Webhooks, HMAC-SHA256, JWT/OAuth, Async processing
AI / APIs: Claude, OpenAI GPT-4o, LLM API integration, Prompt engineering, Structured JSON output
Databases: PostgreSQL (Supabase), Snowflake, SQL, MongoDB
Tools: Git, Docker, AWS (EC2), Apache Airflow, GitHub Actions (CI/CD), GitHub Apps API, Vercel, Render, Power BI, Unix/Linux

EXPERIENCE

Data Analytics Intern – DOWC *June 2026 – Present*
Parsippany, NJ (Onsite)
- Built a reusable Apache Airflow ETL framework for SFTP-to-PostgreSQL pipelines, standardizing DAG architecture, configuration patterns, and shared utilities across production workflows serving major client data feeds
- Leveraged Snowflake's columnar architecture and micro-partitioning to optimize analytical workloads, reducing complex query times from 20+ minutes to under 1 minute on multi-million row datasets
- Reverse-engineered Power BI reporting logic by tracing nested DAX measures and translating calculations into SQL, producing business-facing data lineage documentation for stakeholder validation

Technical Business Analyst Intern – Marketeq Digital *Sept 2024 – Feb 2025*
Miami, FL (Remote)
- Bridged engineering and business stakeholders: translated wireframes into user stories and acceptance criteria, enabling modular feature rollouts across 3 product teams
- Mapped data integration flows across Strapi CMS, MongoDB, and Customer.io on Figma to support personalized automated email campaigns for 150+ clients

PROJECTS

GitHub Review Bot | github.com/pvparekh/github-review-bot *Apr 2026 – May 2026*
- Developed an autonomous GitHub App AI code reviewer using Python, FastAPI, and Claude that processes pull requests and posts structured inline comments within ~10 seconds of a push
- Secured webhook server with HMAC-SHA256 signature verification and RSA JWT authentication; built unified diff parser feeding a two-pass Claude pipeline that returns structured JSON per line and a markdown PR summary
- Containerized and deployed the application using Docker and AWS EC2; configured GitHub Actions CI/CD for automated redeployment workflows

Formula Vision | github.com/pvparekh/F1-Viewer | formulavision.vercel.app *Dec 2025 – May 2026*
- Built a full-stack F1 race replay platform with React, TypeScript, FastAPI, and WebSockets; engineered 60 FPS interpolated animation from 12.5 FPS server data using SVG viewBox transformations and requestAnimationFrame, streaming 440MB of historical telemetry across 5 Grand Prix races
- Optimized memory footprint from 600MB to 150MB for Render’s 512MB free tier through LRU cache eviction with gc.collect(), strategic 30% frame truncation (95MB to 66MB files), and concurrent connection limiting, enabling efficient production deployment
- Integrated FastF1 API for verified race results; implemented dynamic camera follow system with 2.5x zoom tracking, playback controls (play/pause/seek, 4x speed, ±10s skip), real-time grid and driver telemetry panel

AetherFlow: Expense Intel | github.com/pvparekh/aetherflow | aetherflow-three.vercel.app *July 2025 – Apr 2026*
- Architected and shipped a full-stack SaaS-style expense intelligence platform: Next.js 15 App Router, TypeScript, Tailwind CSS, Supabase (PostgreSQL + Auth), deployed to Vercel; supports CSV, PDF, and TXT uploads
- Designed a two-pass AI pipeline: GPT-mini batch-categorizes 25 transactions per call into 9 categories; GPT-4o asynchronously generates a financial health score, insights, and savings opportunities from pre-aggregated statistics
- Built deterministic analytics engine with per-transaction z-scores, MAD z-scores, rolling category averages, drift detection (>15% shift), and vendor intelligence at zero inference cost